

MP2 Report

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Overview

The circuit drives the RGB LED on the iceBlinkPico board so that it travels once around the Hue, Saturation, and Value (HSV) color wheel every 1 second. Each color channel is driven by a PWM signal whose duty cycle follows the piecewise-linear RGB-versus-hue waveforms given in the assignment.

The design targets the iCE40UP5K FPGA clocked by the board's 12 MHz oscillator. It is written in SystemVerilog as three source files: `top`, `color_cycle`, `pwm` and simulated with Icarus Verilog using the test bench file `mp2_tb.sv`.

- Source files and test bench: github.com/leoliao0606-creator/CompArch-Leo/tree/main/MP2
- Video demo on the board: [MP2 Video.mp4](#)

System architecture

The top-level module `top` connects one `color_cycle` module to three identical `pwm` modules, all clocked by the board's 12 MHz oscillator.

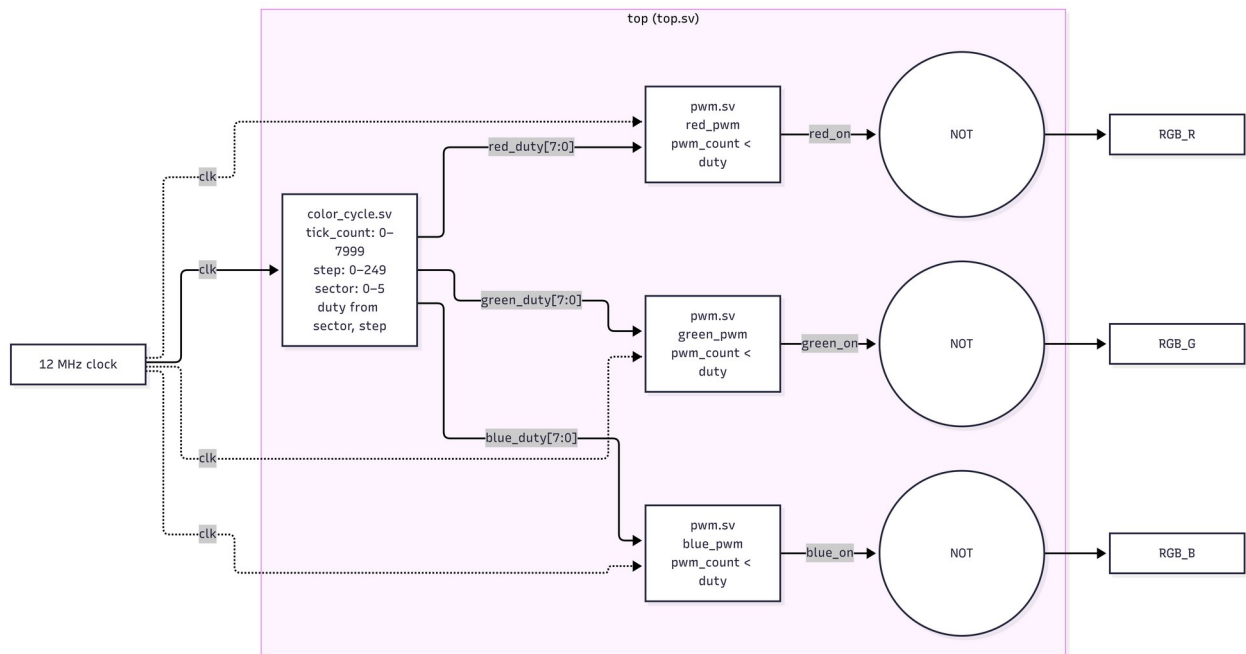


Figure 1. System block diagram.

`color_cycle` produces three 8-bit duty values. Each `pwm` turns one duty value into a PWM signal, and an inverter then drives the matching active-low LED pin. The same `clk` net feeds all four modules, so the three PWM counters stay in phase.

Module descriptions

color_cycle.sv

color_cycle acts like a duty-cycle generator which turns the 12 MHz clock into three 8-bit duty-cycle values that trace the RGB-versus-hue waveforms once per second. Its only input is clk; its outputs are red_duty, green_duty and blue_duty, each in the range 0–250, where 250 means 100% (I did this approximation for).

Three cascaded counters, updated in one always_ff block, track where the system is on the color wheel:

Counter	Width	Range	Advances when
tick_count	13 bits	0–7999	every clock
step	8 bits	0–249	tick_count wraps every 8000 cycles
sector	3 bits	0–5	step wraps every 250 cycles

The parameters CLOCKS_PER_STEP = 8000, STEPS_PER_SECTOR = 250 and NUM_SECTORS = 6 set these limits, so $8000 \times 250 \times 6 = 12,000,000$ clocks make one full revolution.

The always_comb block then maps sector and step onto the three duty values. In each 60° sector one channel holds at 250, one ramps (rising up with step or falling down with 250 – step), and one stays at 0:

sector	Hue	red_duty	green_duty	blue_duty
0	0°–60°	250	step (rising)	0
1	60°–120°	250 – step (falling)	250	0
2	120°–180°	0	250	step (rising)
3	180°–240°	0	250 – step (falling)	250
4	240°–300°	step (rising)	0	250
5	300°–360°	250	0	250 – step (falling)

pwm.sv

As for the pulse-width modulator, the pwm module converts one 8-bit duty value into a 1-bit PWM signal on which determines if the light turn on or not in the current tick. It contains an 8-

bit counter `pwm_count` that counts 0 to 249 and wraps, so one PWM period is 250 clocks. The output is on = (`pwm_count < duty`), which is high for duty clocks out of every 250. A duty of 0 keeps the LED off, and a duty of 250 keeps it on.

The PWM period of 250 clocks was chosen to match the 0–250 duty range from `color_cycle`, so no scaling is needed between the two modules.

top.sv

`top` is the module that coordinates other modules and synthesized onto the FPGA. It takes `clk` from the 12 MHz clock and drives `RGB_R`, `RGB_G` and `RGB_B`. It instantiates one `color_cycle` and three `pwm` modules for each color channel (`red_pwm`, `green_pwm`, `blue_pwm`), and connects each duty output of `color_cycle` to the duty input of one `pwm`.

The RGB LED on the iceBlinkPico is active low: driving a pin low turns its LED on. `top` therefore inverts each PWM signal before it reaches the pin (for example `RGB_R = ~red_on`).

mp2_tb.sv

`mp2_tb` instantiates `top` as `u0` and drives it with a 12 MHz clock. It runs for $3 \times 12,000,000 + 1$ clock cycles, which is three full color-wheel revolutions, and records the three duty-cycle values and the three PWM signals to `mp2.vcd` for viewing in GTKWave.

Simulation results

The simulation shows the duty cycles repeating the required RGB-versus-hue pattern exactly once per second.

The design was simulated with Icarus Verilog for three full color-wheel revolutions, and the resulting `mp2.vcd` was viewed in GTKWave.

Duty-cycle waveforms

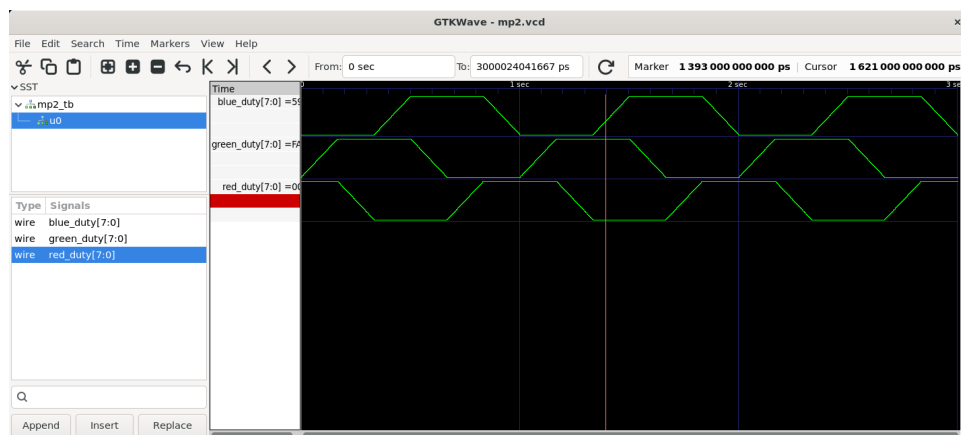


Figure 2. GTKWave plot of `blue_duty`, `green_duty` and `red_duty` (8-bit values, analog step format) over three color-wheel revolutions.

Figure 2 shows three full revolutions. Each channel is a trapezoid that repeats once per revolution: it ramps up for one sector, holds at 250 for two sectors, ramps down for one sector

and holds at 0 for two sectors. Green lags red by two sectors (120°) and blue lags green by another two sectors, matching the assignment's waveforms. At $t = 0$ red is at 250 and green starts rising from 0, which is hue 0° (red).

PWM signals

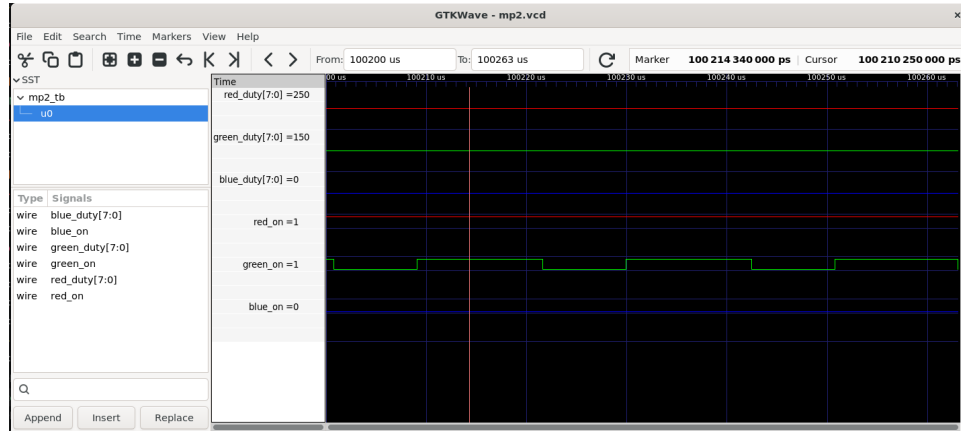


Figure 3. GTKWave plot of the duty cycles and the PWM outputs red_on, green_on and blue_on over three PWM periods.

In this window red_duty is 250, so red_on stays high. green_duty is 150, so green_on is high for 150 of every 250 clocks (60%). blue_duty is 0, so blue_on stays low.